

CHAPTER 1 — HOW NEXT.JS UI WORKS

1. Components:

A component is a reusable building block of UI. Navbar, buttons, cards—all components.

2. layout.js:

This wraps every page. It defines global structure like navbar, footer, metadata.

3. Why Navbar only once:

layout.js renders on every page, so adding navbar here ensures no duplicates.

4. page.js:

This is the actual page content shown inside layout.js.

5. Tailwind CSS:

Utility-based CSS. You add classes like p-4, bg-black to style instantly.

6. globals.css:

Global styles affecting the entire project. Good for custom CSS and resets.

CHAPTER 2 — IMAGES & LAYOUTS

1. Why images appear huge:

Images expand to container width. If container is large, image looks large.

2. Grids:

CSS Grid arranges items in rows/columns.

3. Masonry:

Pinterest-style waterfall layout. Needs break-inside-avoid and columns.

4. break-inside:

Prevents item from splitting between columns.

5. object-cover:

Ensures image fills its box without distortion.

CHAPTER 3 — REACT BASICS

1. `projects[]` array:

Holds multiple items. Each item is shown via `.map()`.

2. `.map()`:

Loops through array and returns JSX for each item.

3. JSX:

HTML written inside JavaScript.

4. Props:

Data passed from one component to another.

5. State:

Variables that trigger UI updates when changed.

CHAPTER 4 — ARCHITECTURE UI CONCEPTS

1. Hero sections:

Large banners with text overlay. Usually absolute-positioned.

2. Positioning:

absolute, relative control placement of elements.

3. Gradients:

Used to improve text readability over images.

4. Responsive design:

Tailwind breakpoints like sm, md, lg adjust layout based on screen size.

CHAPTER 5 — BACKEND (SPRING BOOT + MONGO)

1. Model/Document:

Java class representing Mongo DB document.

2. Repository:

Interface that allows CRUD operations.

3. Controller:

Handles API requests (GET, POST, DELETE).

4. Service:

Business logic layer (optional but recommended).

5. JSON & POJO:

Data format and Java objects that map to it.

CHAPTER 6 — FULL SYSTEM ARCHITECTURE

Frontend → API → Backend → Database

1. User interacts with React.
2. React calls Spring Boot API.
3. API saves/retrieves data from MongoDB.
4. Response goes back to frontend.